

SEMESTER-VIII

UME735: MACHINING SCIENCE				
	L	T	P	Cr
	2	1	2	3.5
Course Objective: This course inculcates specialized knowledge and skill in various machining processes using the principles and methods of engineering analysis and design. This course also cultivates the ability to develop and optimize the conventional machining processes resulting in creation and distribution of value in engineering applications.				
Syllabus Machining with Single Point Cutting Tool: Strain rate; Velocity relationships; Estimation of cutting forces and power using various theories as Lee and Shaffer, Palmer and Oxley (Flip learning), Ernst Merchant; Power and energy relationships; Frictional and thermal aspects of machining. Machining with Multi Point Cutting Tools: Analysis of chip cross-section; Determination of forces and power requirement in various types of milling, drilling and grinding operations. Tool Failure: Types of tool failure; factors influencing surface quality, dimensional accuracy and material removal rate in machining; Cutting fluids; Calculation of economic cutting speed; Gilberts model for economic tool life (Flip learning); Determination of optimal cutting speed for maximum production; Economics of multistage cutting; high efficiency zone. Jigs and Fixtures: Elements and importance of jigs and fixtures in machining; Design considerations for jigs and fixtures; Materials used for jigs and fixtures; Principle of Location; Locating methods; Jigs and their various types; fool proofing; Types of Fixtures; Methods of Clamping; Mechanical, pneumatic, hydraulic and vacuum clamping; Case studies on relevant topics (Flip learning).				
Laboratory Work Experimental work pertaining to determination of chip reduction coefficient, shear angle; Cutting force measurements in milling and drilling operations; Effects of speed, feed and depth of cut on power consumption; tool tip temperature. Research Assignment: Students will be divided into groups of 4–5 students. Each group will be assigned with a separate research topic related to parametric analysis and optimization of process parameters involved in various machining processes. Students will be required to go through the topics and recent developments from sources like reference books, journals etc. in the relevant field. Each group will be required to submit a report (and presentation) containing review of literature, summary, gaps in the existing literature, key findings etc.				
Course Learning Outcomes (CLOs) The students will be able to: 1. calculate cutting forces and power requirements during single point and multi-point cutting operations. 2. analyze the thermal and frictional aspects of machining processes. 3. design the conditions for the maximum tool life and factors influencing surface quality, dimensional accuracy and material removal rate in machining.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

4. select the various types of jigs and fixtures to be used for different machining processes.

Text Books

1. Ghosh, A., Mallik, A.K., Manufacturing Science, Affiliated East-West Press Pvt. Ltd, New Delhi (2013).
2. Bhattacharyya , A., Metal Cutting Theory and Practice, New Central Book Agency Ltd, Calcutta (2000).
3. Pandey, P. C., Singh, C. K., Production Engineering Sciences, Standard Publishers, New Delhi (2004).
4. Mehta, N., Metal Cutting and Design of Cutting Tools, Jigs & Fixtures, McGraw Hill Education (India) Private Limited, New Delhi (2015)
5. Rao, P.N., Manufacturing Technology Vol 2 : Metal Cutting & Machine Tools, McGraw Hill Education (2018)

Reference Books

1. Juneja, B. L., Sekhon, G. S., Metal Cutting, New Age International, New Delhi (2003).
2. Shaw, M.C., Metal Cutting, Tata McGraw Hill, New Delhi (1997).

Evaluation Scheme:

S N	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessionals (Assignments/Projects/Tutorials/Quizzes)	25

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